

INTERNSHIP PROJECT REPORT

Data Analytics Internship • Codec Technologies India

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Program / Domain	Data Analytics Internship
Organization	Codec Technologies India
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1. Summary

This report covers the first month (18 May – 18 June 2026) of a two-month Data Analytics internship at Codec Technologies India. The internship follows a structured curriculum spanning the full data analytics lifecycle: introduction to data analytics, data cleaning and preprocessing, exploratory data analysis (EDA), advanced analysis, machine learning, big data and cloud computing, and a final industrial project.

During this period, the foundational modules were completed and the concepts were applied by building the first hands-on project — a Customer Churn Prediction model that predicts which telecom customers are likely to stop using a service. The project was developed as a complete Jupyter notebook covering data cleaning, EDA, three classification models, and a full evaluation, achieving a ROC-AUC of approximately 0.84.

2. Tasks Completed

2.1 Coursework Modules

- **Introduction to Data Analytics** — the analytics lifecycle, the four types of analytics, and the standard toolset (Excel, SQL, Python, R, Power BI / Tableau).
- **Data Cleaning & Preprocessing** — handling missing/invalid values, type conversion, encoding, and feature scaling.
- **Exploratory Data Analysis (EDA)** — summary statistics, distribution and correlation analysis, and visual pattern identification.
- **Machine Learning Fundamentals** — supervised classification, train/test split, model fitting, and evaluation metrics.

2.2 Project 1 — Customer Churn Prediction

Objective: Predict which telecom customers are likely to churn, so the business can target them with retention offers.

Dataset: Telco Customer Churn — 7,043 customers with 21 attributes (demographics, services, contract type, billing).

- Cleaned the data (converted the TotalCharges text field to numeric and handled blanks for new customers).
- Performed EDA to uncover churn patterns by tenure, contract type, and monthly charges.

- Encoded features, applied a stratified 80/20 split, and trained Logistic Regression, Random Forest, and XGBoost with class balancing.
- Evaluated using accuracy, recall, ROC-AUC, ROC curves, and a confusion matrix, then ranked the key churn drivers.

Result: Best ROC-AUC ≈ 0.84 . The strongest churn drivers were short tenure, month-to-month contracts, and high monthly charges — a clear, actionable signal for a retention campaign.

3. Challenges Faced

- **Messy real-world data.** A numeric field was stored as text with blank entries, which had to be detected and corrected before modelling.
- **Class imbalance.** Churners were the minority, so a naive model could score high accuracy while missing them; this was handled with class-balancing.
- **Choosing the right metric.** Accuracy alone was misleading, so the focus shifted to recall and ROC-AUC.
- **Environment setup.** Installing and configuring libraries such as XGBoost and producing a clean, reproducible notebook took some initial troubleshooting.

4. Learning and Skills Gained

- **Technical:** Python (pandas, NumPy), data visualisation (Matplotlib, Seaborn), and machine learning (scikit-learn, XGBoost) in documented Jupyter notebooks.
- **Analytical:** the end-to-end analytics workflow, feature encoding and scaling, stratified splitting, classifier evaluation (accuracy, recall, ROC-AUC), and handling imbalanced data.
- **Professional:** translating model output into business recommendations and writing clear technical documentation.

5. Plan for the Remaining Days of the Internship

Planned work for the second month (18 June – 18 July 2026):

Module / Phase	Planned Activity
Project 2 — Diabetes Risk Detection	Healthcare classification model (Pima dataset) with feature-importance analysis and a data-privacy note.
Advanced & Applied ML	Hyperparameter tuning, cross-validation, and model interpretability (SHAP).
Big Data & Cloud Computing	Processing data at scale and using cloud platforms for analytics.
Final Industrial Project	Deliver the final end-to-end industrial project with a dashboard.